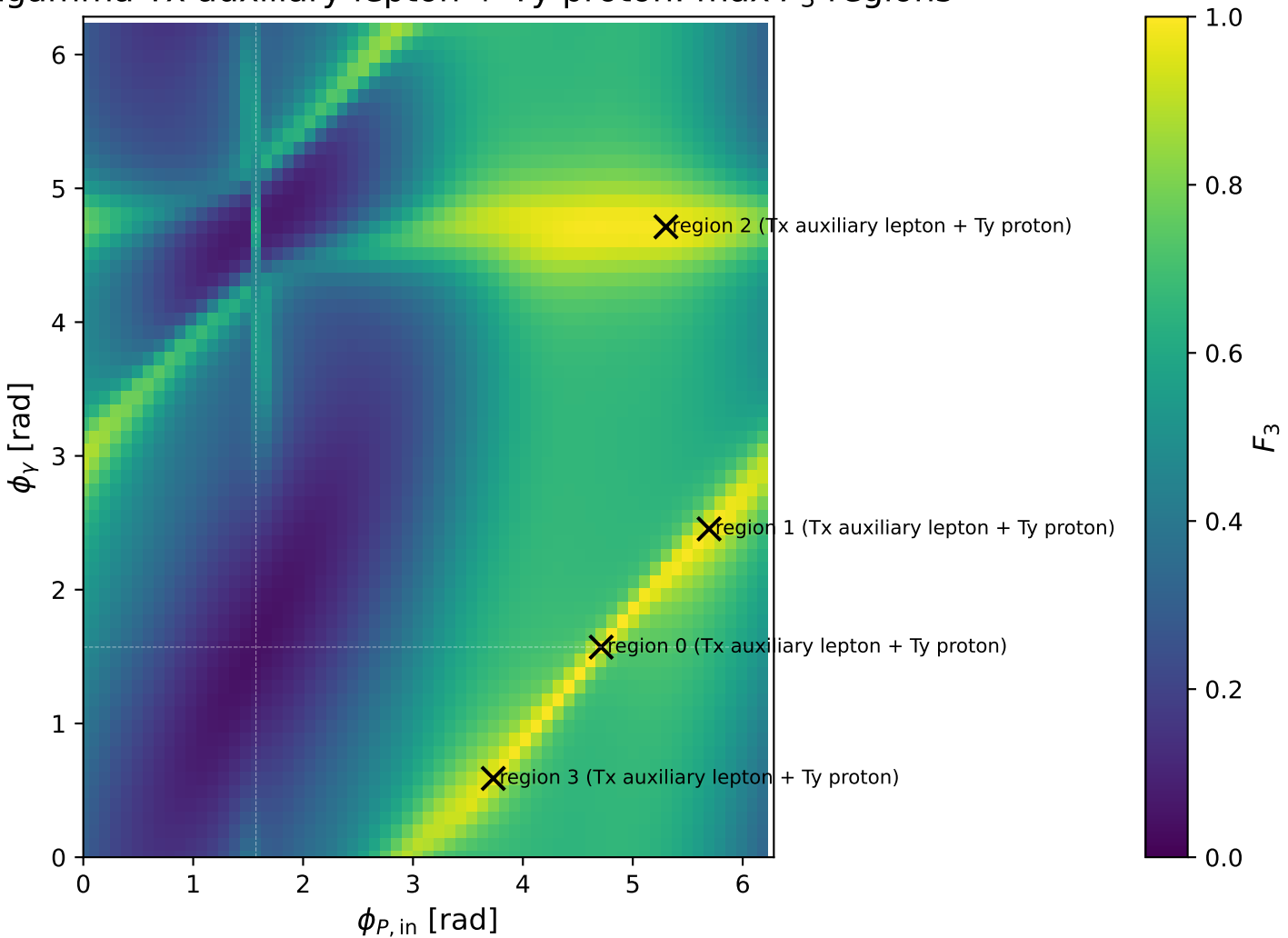
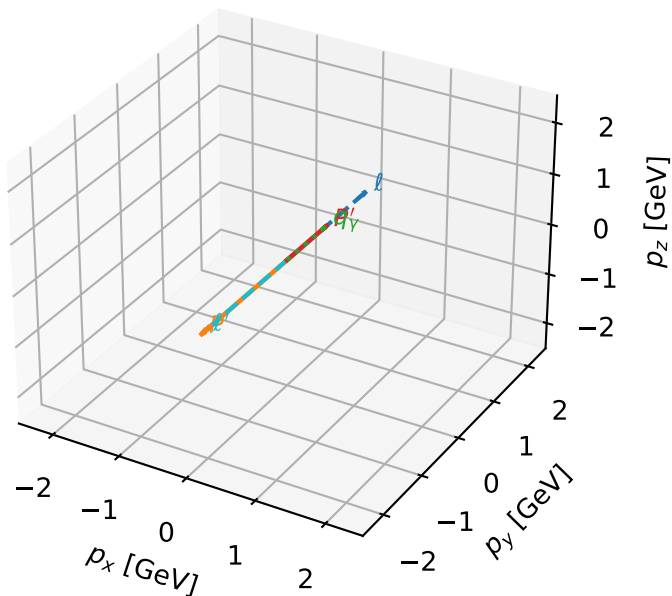


gamma Tx auxiliary lepton + Ty proton: max F_3 regions



Tx auxiliary lepton + Ty proton characteristic max F_3 configuration

3D momenta

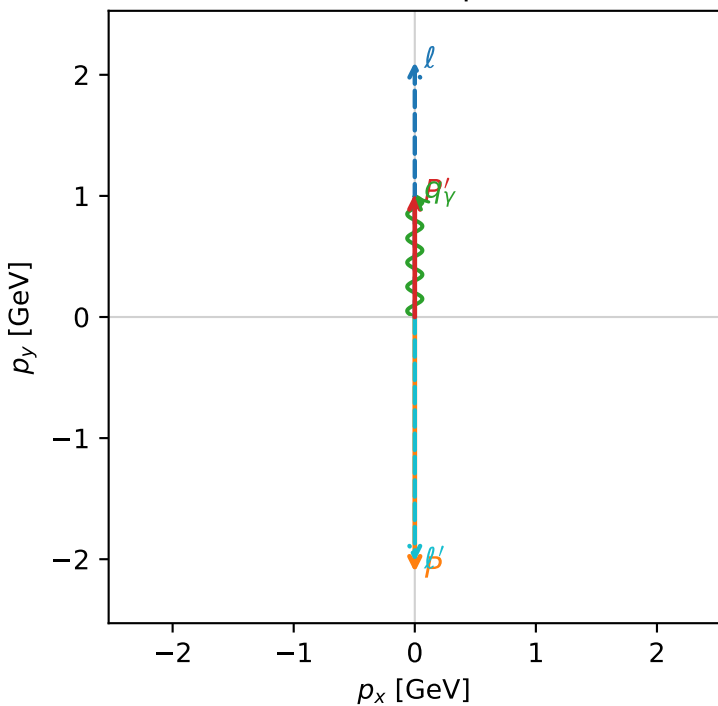


f3_mid_egamma_txtty_region_0 $F_3=0.996287$
 region: mid_Egamma_TxTy_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|T_{x,\ell}\rangle \otimes |\bar{T}_{y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.01927$
 $\theta_{in}=1.57079$, $\phi_\ell=1.5708$, $\phi_p=4.71239$
 $E_\gamma=1$, $\phi_\gamma=1.5708$
 $Q^2=17.0194$, $x_B=0.958772$, $t=-8.92478$, $W^2=1.61168$, $y=0.904017$
 $\theta(\ell', \gamma)=180$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= 0.0000$ $p_y= 2.1069$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -0.0000$ $p_y= -2.1069$ $p_z= 0.0000$
 ℓ' $E= 2.2533$ $p_x= -0.0000$ $p_y= -2.0193$ $p_z= 0.0000$
 P' $E= 1.3852$ $p_x= 0.0000$ $p_y= 1.0193$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.0000$ $p_y= 1.0000$ $p_z= 0.0000$

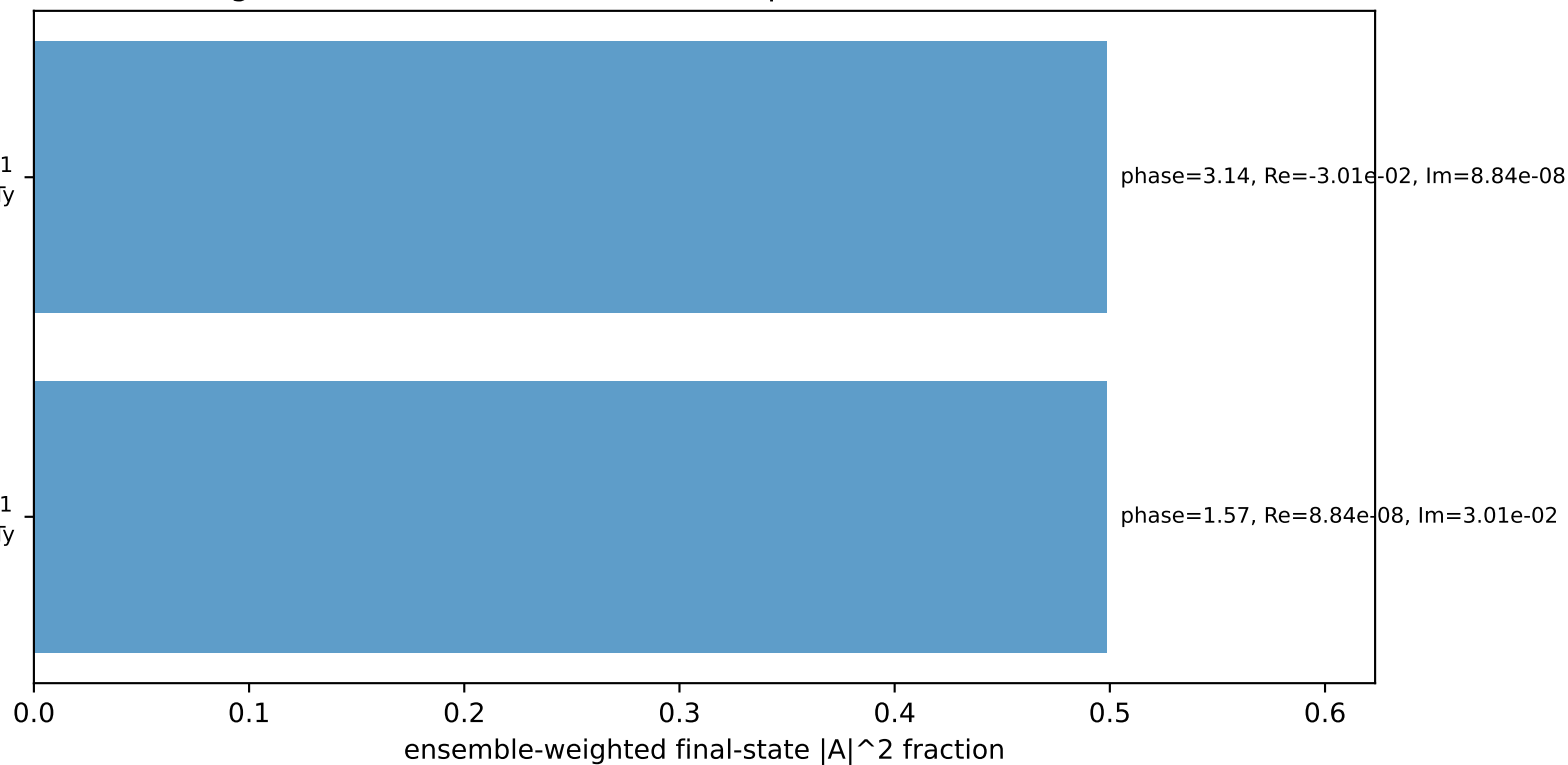
Transverse plane



$h_e=-1$, $h_p=-1$, $h_\gamma=+1$
 electron Tx, proton Ty

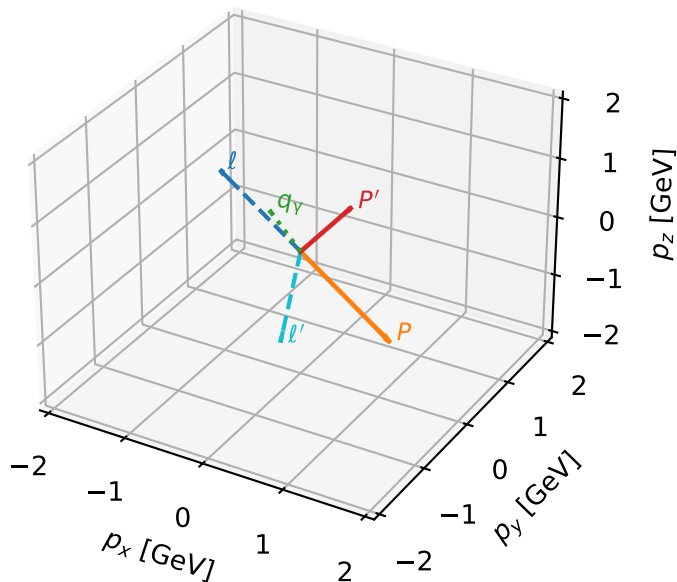
$h_e=+1$, $h_p=+1$, $h_\gamma=-1$
 electron Tx, proton Ty

Leading initial-ensemble/final-state components (N=2, retained=99.7%)



Tx auxiliary lepton + Ty proton characteristic max F_3 configuration

3D momenta

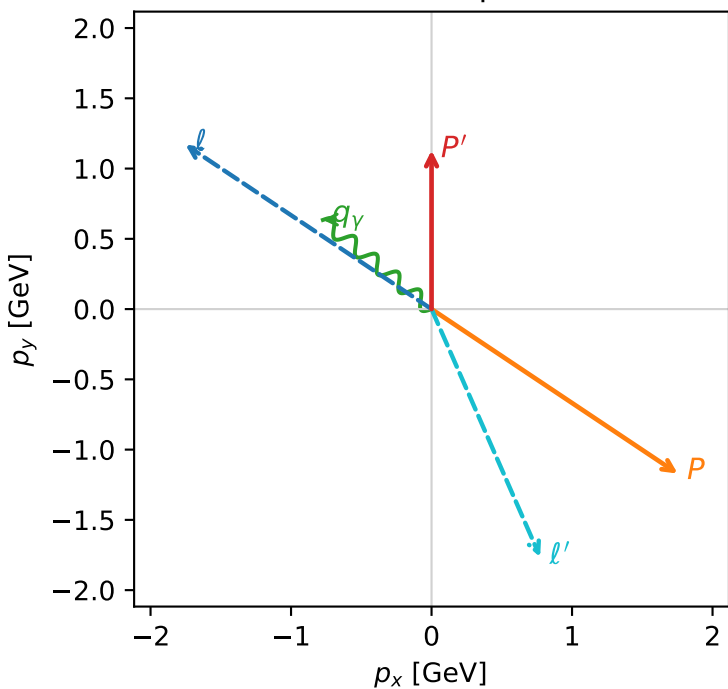


f3_mid_egamma_txtty_region_1 $F_3=0.993983$
 region: mid_Egamma_TxTy_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|T_{x,\ell}\rangle \otimes |\bar{T}_{y,p}\rangle$

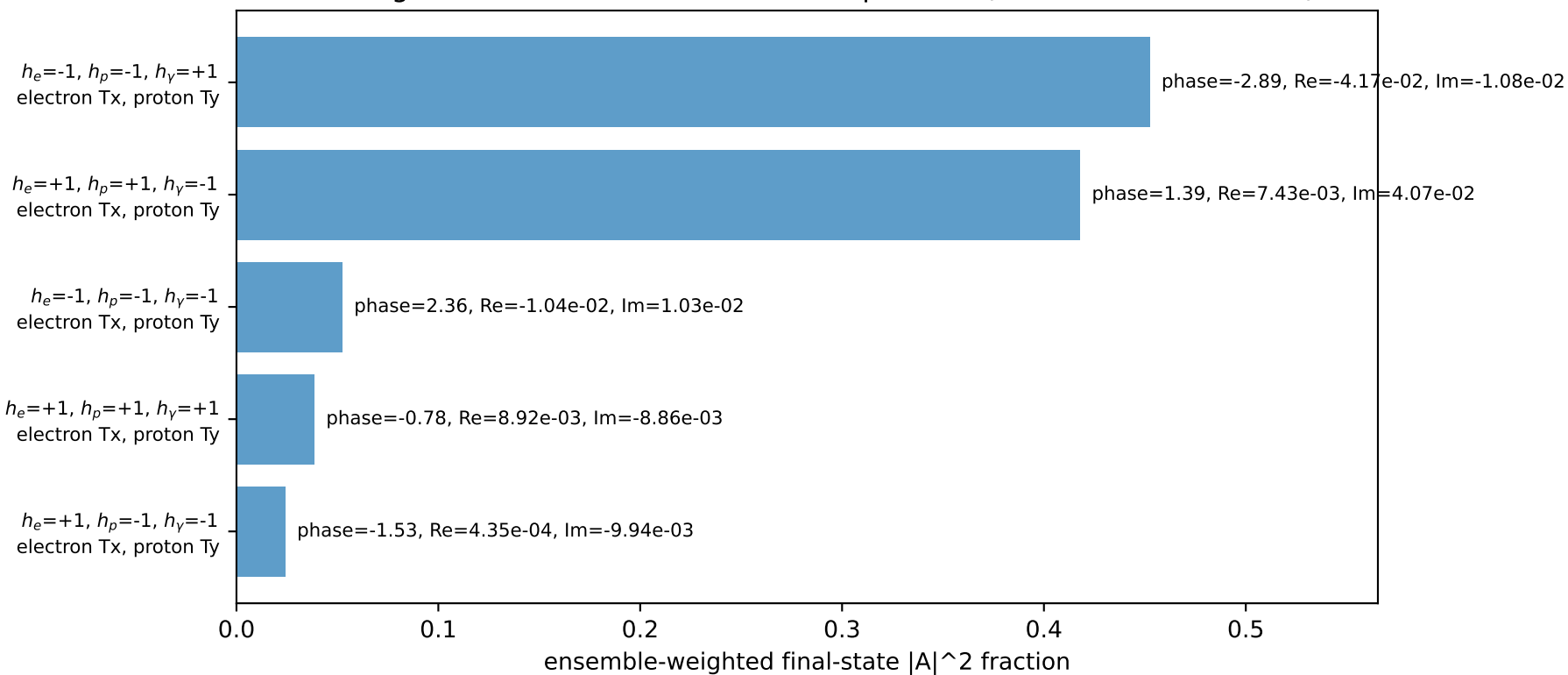
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.12971$
 $\theta_{in}=1.57079$, $\phi_l=2.55254$, $\phi_P=5.69414$
 $E_\gamma=1$, $\phi_\gamma=2.45437$
 $Q^2=14.9609$, $x_B=0.908689$, $t=-7.65805$, $W^2=2.38321$, $y=0.838473$
 $\theta(\ell', \gamma)=153.038$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= -1.7519$ $p_y= 1.1706$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 1.7519$ $p_y= -1.1706$ $p_z= 0.0000$
 ℓ' $E= 2.1702$ $p_x= 0.7730$ $p_y= -1.7641$ $p_z= 0.0000$
 P' $E= 1.4684$ $p_x= 0.0000$ $p_y= 1.1297$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.7730$ $p_y= 0.6344$ $p_z= 0.0000$

Transverse plane

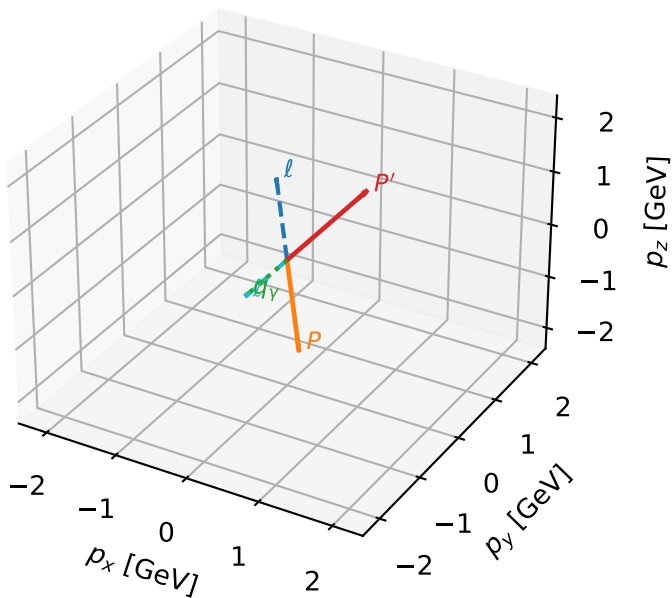


Leading initial-ensemble/final-state components (N=5, retained=98.5%)



Tx auxiliary lepton + Ty proton characteristic max F_3 configuration

3D momenta

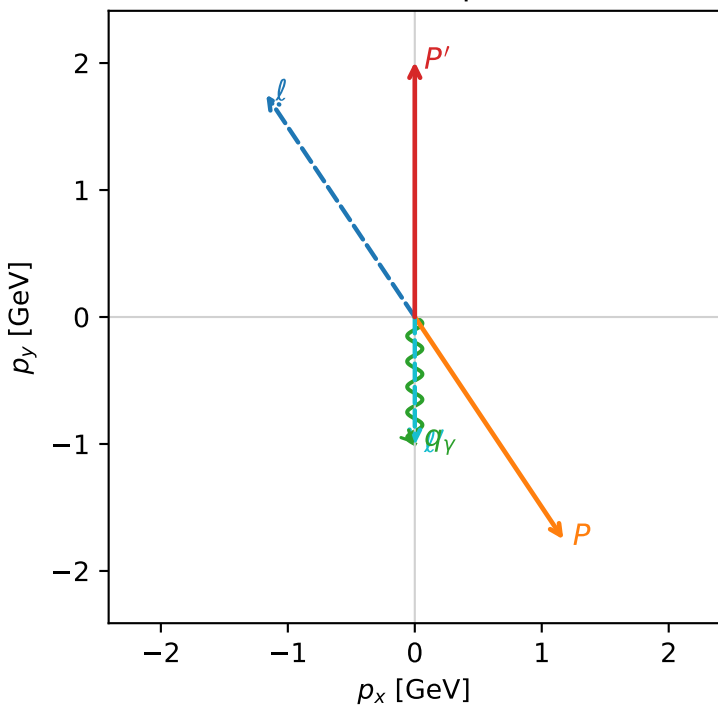


f3_mid_egamma_txtty_region_2 $F_3=0.989138$
 region: mid_Egamma_TxTy_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|T_{x,l}\rangle \otimes |\bar{T}_{y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=2.00946$
 $\theta_{in}=1.57079$, $\phi_l=2.15984$, $\phi_p=5.30144$
 $E_Y=1$, $\phi_Y=4.71239$
 $Q^2=8.1646$, $x_B=0.491289$, $t=-15.5098$, $W^2=9.33396$, $y=0.846339$
 $\theta(l', \gamma)=0$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= -1.1706$ $p_y= 1.7519$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 1.1706$ $p_y= -1.7519$ $p_z= 0.0000$
 l' $E= 1.4209$ $p_x= 0.0000$ $p_y= -1.0095$ $p_z= 0.0000$
 P' $E= 2.2176$ $p_x= 0.0000$ $p_y= 2.0095$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= -0.0000$ $p_y= -1.0000$ $p_z= 0.0000$

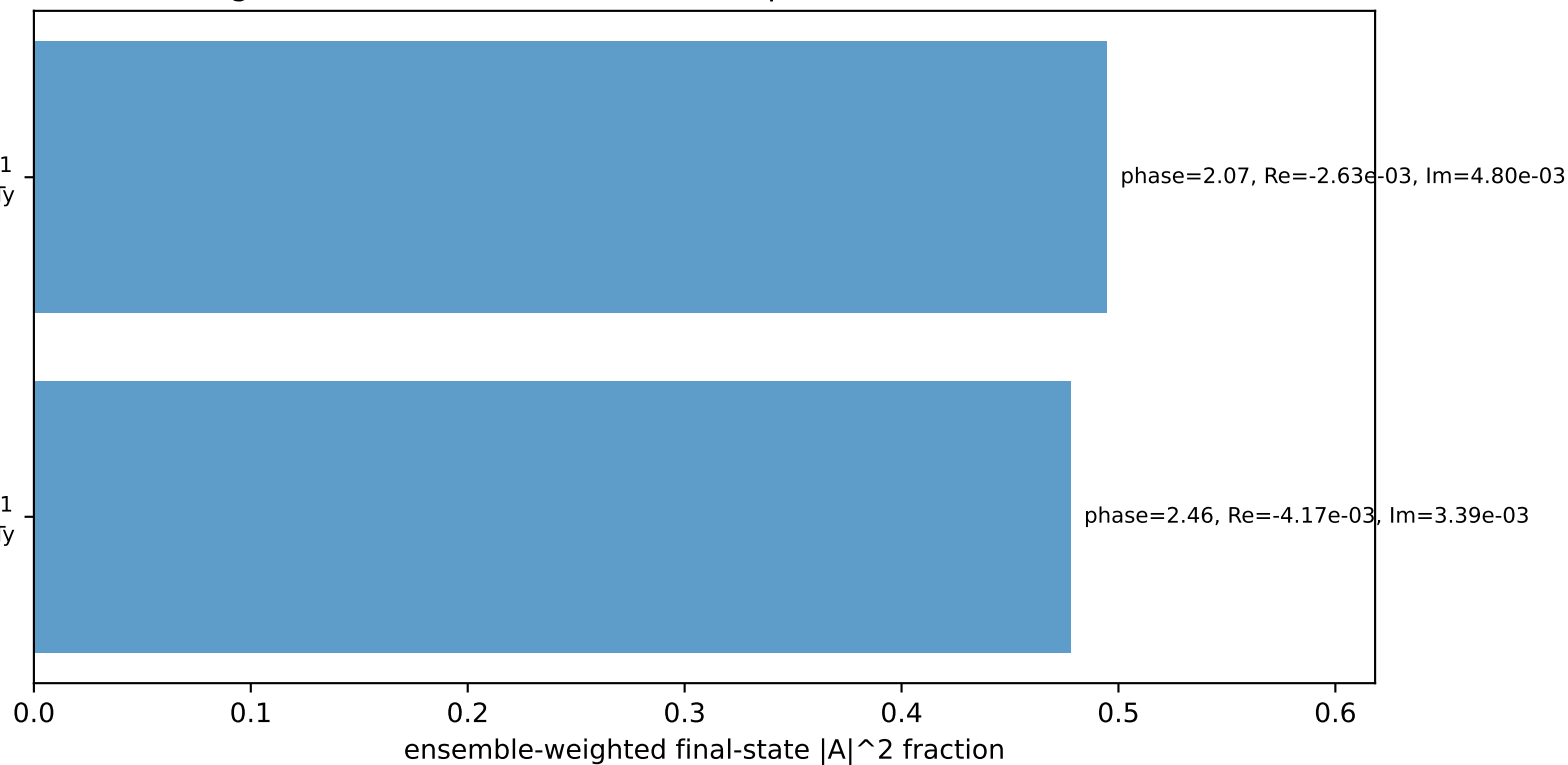
Transverse plane



$h_e=-1$, $h_p=+1$, $h_Y=+1$
 electron Tx, proton Ty

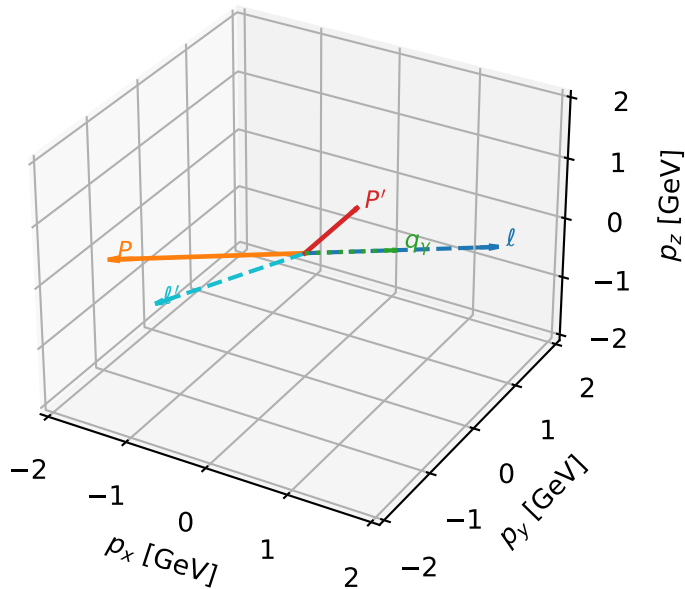
$h_e=+1$, $h_p=-1$, $h_Y=-1$
 electron Tx, proton Ty

Leading initial-ensemble/final-state components (N=2, retained=97.3%)



Tx auxiliary lepton + Ty proton characteristic max F_3 configuration

3D momenta

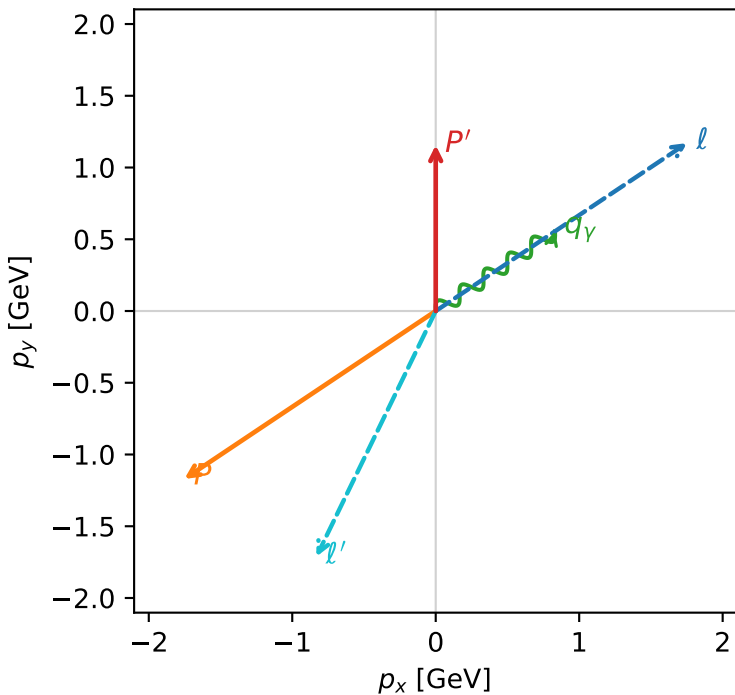


f3_mid_egamma_txtty_region_3 $F_3=0.97462$
 region: mid_Egamma_TxTy_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|T_{x,l}\rangle \otimes |\bar{T}_{y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.15614$
 $\theta_{in}=1.57079$, $\phi_l=0.589049$, $\phi_p=3.73064$
 $E_Y=1$, $\phi_Y=0.589049$
 $Q^2=14.9478$, $x_B=0.898264$, $t=-7.81421$, $W^2=2.5728$, $y=0.847459$
 $\theta(l', \gamma)=149.658$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= 1.7519$ $p_y= 1.1706$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -1.7519$ $p_y= -1.1706$ $p_z= 0.0000$
 l' $E= 2.1497$ $p_x= -0.8315$ $p_y= -1.7117$ $p_z= 0.0000$
 P' $E= 1.4888$ $p_x= 0.0000$ $p_y= 1.1561$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= 0.8315$ $p_y= 0.5556$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=3, retained=96.9%)

